

Aether One™ – RVV 1.0 Toolchain & TVM Optimization Guide

Preparation and configuration of RISC-V RVV 1.0 toolchain and TVM optimization workflow for TrustCore NX acceleration.

Purpose

To unlock the 32 TOPS potential of the TrustCore NX processor, the Aether One development stack must support RISC-V Vector Extension (RVV 1.0). This guide provides step-by-step instructions to build the GNU toolchain with RVV support and to integrate RVV-optimized targets into Apache TVM.

System Overview

Aether One Build

Environment ■■■■■■
Host (x86 / Pi 5 Linux) ■■■■ • Build RISC-V GNU Toolchain (RVV 1.0) ■■■■ • Cross-compile TVM models for TrustCore NX ■■■■
TrustCore NX
(Target SoC) ■■■■ • Runs Aether OS (Linux) + libtvm_runtime.so ■■■■ • Executes RVV-optimized inference binaries ■■■■
■■■■■

Phase 1 – Build RISC-V GNU Toolchain with RVV 1.0 Support

Use the official `riscv/collab/riscv-gnu-toolchain` repository to compile a cross-compiler supporting RVV 1.0.

```

sudo apt update
sudo apt install -y autoconf automake libmpc-dev libmpfr-dev libgmp-dev \
gawk build-essential bison flex texinfo gperf libtool patch python3 \
mkdir -p $HOME/riscv/toolchain
cd $HOME/riscv/toolchain
git clone --recursive https://github.com/riscv-collab/riscv-gnu-toolchain
cd riscv-gnu-toolchain
export RISCV_TOOLCHAIN_INSTALL=$HOME/riscv/toolchain/install
./configure --prefix=$RISCV_TOOLCHAIN_INSTALL \
--with-arch=rv64gcv --with-abi=lp64d --enable-multilib
make -j$(nproc)
linux echo 'export PATH="$HOME/riscv/toolchain/install/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
The resulting binaries (riscv64-unknown-linux-gnu-gcc / g++) support RVV 1.0 (-march=rv64gcv).

```

Phase 2 – Integrate RVV Support into Apache TVM

With the toolchain installed, configure TVM to emit RVV instructions for TrustCore NX targets.

```
import tvm
from tvm.target import Target

target_riscv_rvv = Target(
    "llvm -mtriple=riscv64-unknown-linux-gnu "
    "-mcpu=generic-rv64 -mabi=lp64d "
    "-mattr=+v,+zfh" # Vector + half precision
)

# Example build process
# mod, params = relay.frontend.from_onnx(model, shape_dict)
# with tvm.transform.PassContext(opt_level=3):
#     lib = relay.build(mod, target=target_riscv_rvv, params=params)
# lib.export_library("terra_optimized_rvv.tar")
```

Optimization Notes:

- +v enables RVV 1.0 vector instructions.
- +zfh adds half-precision float support for AI inference.

- Optional flags: +zvl512b, +zvl256b depending on TrustCore NX vector length.
- Use -mno-strict-align if unaligned memory accesses are permitted for higher throughput.

Phase 3 – Deployment and Validation

1. Cross-compile the TVM modules on your host system or Pi 5.
2. Transfer the RVV-optimized library (e.g. terra_optimized_rvv.tar) to the TrustCore NX target.
3. Link it using Aether OS runtime (libtvm_runtime.so).
4. Measure inference throughput and power efficiency to fine-tune RVV vector parameters.

Summary

The RVV 1.0 toolchain and TVM integration enable Aether One to fully leverage TrustCore NX's parallel vector processing capabilities. This is a critical milestone toward hardware-accelerated, post-quantum-secure edge AI performance within the Aether One prototype stack.